Biology Methods Literature RAG

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Purpose and intended demonstration

Retrieve research-method notes, draft a source-linked answer and keep observed evidence separate from interpretation. This is a computational biology learning project with synthetic or fictional inputs, not a clinical or experimentally validated system.

How the implementation works

Use BM25 over short project-authored method notes carrying links to primary sources. Live mode asks a local LLM for a structured answer and citations; replay mode validates a supplied answer.

Evaluation and evidence

Test unknown source rejection and unsupported questions. Citation identifiers must belong to the retrieved context, and quote spans must occur literally in that source note.

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Setup and commands

Requires Python 3.10 or newer and NumPy. Create a virtual environment: python -m venv .venv. On Windows use .venv\Scripts\python.exe if activation is unavailable. Install with python -m pip install -r requirements.txt. Run python app.py, then open results/report.html in your browser. Run python -m unittest discover -v to execute tests. The default seed is 42; change it with --seed. Replace sample_input.json using --input path-to-file.json, preserving its schema. Algorithm-only projects generate synthetic arrays from the seed rather than consuming external files. For actual language-model generation, install and start Ollama, install a model appropriate for your hardware, then run python app.py --mode live --model YOUR_INSTALLED_MODEL. The local endpoint is http://127.0.0.1:11434/api/generate. A model download and additional memory are required; none were installed or tested for this build. Live requests time out after 120 seconds and failures are not silently replaced by fixture output.

Architecture and file responsibilities

CLI arguments -> JSON input or seeded generator -> domain pipeline -> validation or evaluation -> JSON and HTML artifacts. Imported functions can be used in a notebook or a separate application. Model checkpoints preserve learned arrays where applicable; see app.py for parameter order. This package does not include a production web service. Each project is independent and includes its own helper code so the ZIP can be extracted and run alone. app.py holds the experiment and domain functions: validate, run. kit.py contains reporting, input loading and only the retrieval or generation helpers needed by this project. sample_input.json contains editable inputs. test_app.py checks domain behaviors; test_reporting.py checks metrics and HTML output boundaries. results/metrics.json records outputs and results/report.html provides a browser-readable report. Numeric projects also save arrays or learned parameters to results/artifacts.npz; inspect its keys with numpy.load. MODEL_CARD.md explains intended use and limits. .github/workflows/tests.yml runs tests and the fixture experiment. No API key is needed.

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Limitations and next improvements

Notes are brief paraphrases rather than a comprehensive literature corpus. Citation validation does not prove scientific entailment or consensus, and generated answers are not medical advice. Ingest licensed full texts with versioned provenance, compare lexical and semantic retrieval, label claim support and evaluate contradictory sources.

Interview walkthrough

Present the problem and demonstrate one normal example, then one failure or boundary case. Explain the baseline or fixture mode, the split or validation design, and one limitation before discussing an extension. Be ready to trace data through the functions listed above. Explain why the chosen metric matches the task and why the available results cannot establish production performance. Add your own documented experiment and be precise about what you implemented or changed. These are AI-assisted educational projects; filenames and synthetic scores are not evidence of client work.

Research context and reading

These topics were selected from research directions reviewed on 22 October 2026. Recent activity suggests useful skills to develop; it is not a forecast of hiring, commercial success or scientific validity. The implementation is a small educational baseline and does not reproduce or claim the performance of any cited foundation model. Reading: Universal cell embedding provides a foundation model for cell biology (2026): https://www.nature.com/articles/s41586-026-10689-z; Benchmarking algorithms for generalizable single-cell perturbation response prediction (2025/2026): https://www.nature.com/articles/s41592-025-02980-0.

Verified build results

This build verified fixture replay and validation only. No LLM weights were downloaded and live generation quality was not measured.

project

"Biology Methods Literature RAG"

mode

"fixture"

response

{"answer": "Use independent donor or study holdouts and compare with simple baselines.", "evidence": [{"source": "cell-representations", "quote": "A practical benchmark should hold out donors or studies, compare simple baselines, and distinguish cell annotation from biological mechanism discovery."}]}

source links

{"cell-representations": "https://www.nature.com/articles/s41586-026-10689-z", "perturbation-evaluation": "https://www.nature.com/articles/s41592-025-02980-0", "microscopy-evaluation": "https://www.nature.com/articles/s41592-025-02879-w"}

scope

"Research notes and provenance checks; no clinical guidance"

See results/metrics.json for all observations and TEST_RESULTS.txt for the executed tests. All datasets and replay responses shipped here are synthetic. Results can vary across seeds and numerical library versions.

Provider references: https://docs.ollama.com/api/generate and https://docs.ollama.com/capabilities/structured-outputs. The adapter uses non-streaming generation and requests JSON schema output where applicable. Independent application validation still runs after generation.